SMART PARKING FEE MANAGEMENT

Python Case Study

1. Problem Statement

Manual parking fee calculation can lead to mistakes when calculating the parking duration and fee for different vehicle types. This case study uses a simple Python program to calculate parking hours and the fee for a bike or car based on the entry and exit hours.

2. Objective

To develop a beginner-friendly Python program that accepts the vehicle type, entry hour, and exit hour; validates the input; calculates the parking duration; and displays the parking fee.

3. Assumptions

• Entry and exit times are entered as whole hours from 0 to 23.

• If the exit hour is earlier than the entry hour, the vehicle is assumed to have stayed overnight.

• Bike parking costs Rs. 10 per hour and car parking costs Rs. 20 per hour.

• The fee is capped at Rs. 100.

• The vehicle type must be entered exactly as **bike** or **car**.

4. Algorithm

1. Start the program.

2. Read the vehicle type, entry hour, and exit hour.

3. Check whether the entry and exit hours are between 0 and 23.

4. If either time is invalid, display “Invalid time”.

5. Otherwise, calculate the parking duration, considering overnight parking when needed.

6. If the vehicle is a bike, calculate the fee at Rs. 10 per hour.

7. If the vehicle is a car, calculate the fee at Rs. 20 per hour.

8. If the vehicle type is invalid, display “Invalid vehicle type”.

9. If the calculated fee is greater than Rs. 100, set it to Rs. 100.

10. Display the parking hours and parking fee, then stop.

5. Flowchart

START
↓
Input vehicle type, entry hour, exit hour
↓
Are entry and exit hours within 0–23?
  No → Display “Invalid time” → END

Yes ↓
Is exit hour less than entry hour?
Yes → hours = (24 – entry) + exit
No → hours = exit – entry
↓
Vehicle type?
Bike → fee = hours × 10
Car → fee = hours × 20
Other → Display “Invalid vehicle type”
↓
If fee > 100, set fee = 100
↓
Display parking hours and fee
↓
END

6. Pseudocode

```
BEGIN
    INPUT vehicle
    INPUT entry
    INPUT exit

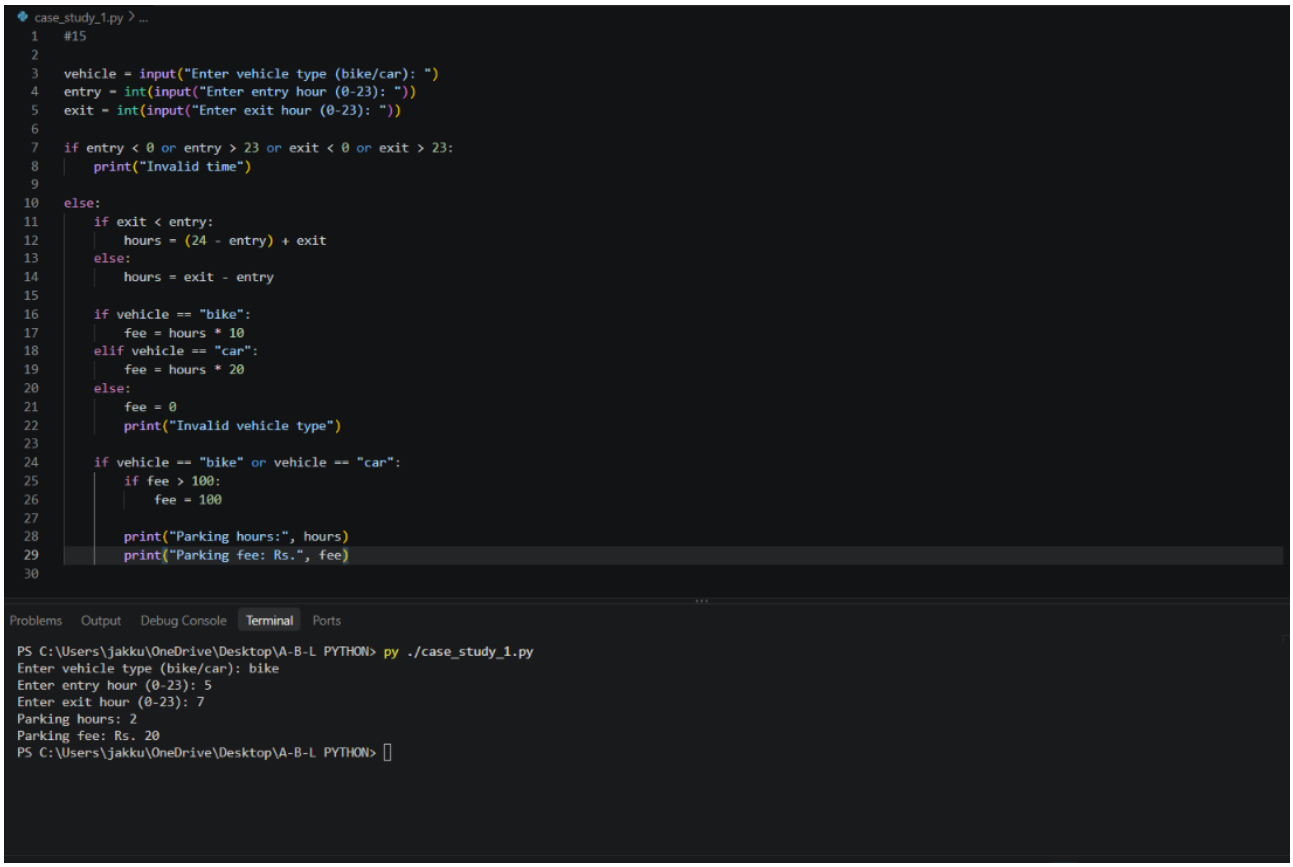
    IF entry < 0 OR entry > 23 OR exit < 0 OR exit > 23 THEN
        PRINT "Invalid time"
    ELSE
        IF exit < entry THEN
            hours = (24 - entry) + exit
        ELSE
            hours = exit - entry
        ENDIF

        IF vehicle = "bike" THEN
            fee = hours * 10
        ELSE IF vehicle = "car" THEN
            fee = hours * 20
        ELSE
            fee = 0
            PRINT "Invalid vehicle type"
        ENDIF

        IF vehicle = "bike" OR vehicle = "car" THEN
            IF fee > 100 THEN
                fee = 100
            ENDIF
            PRINT "Parking hours:", hours
            PRINT "Parking fee: Rs.", fee
        ENDIF
    ENDIF
END
```

7. Python Source Code and Output

Screenshot provided by the student (source code in the upper section and program output in the terminal below).



8. Conclusion

The Smart Parking Fee Management case study demonstrates how Python input statements, conditional statements, and arithmetic operations can be used to calculate parking duration and fees. It also shows basic input validation and a maximum fee limit, making it a useful beginner-level example of solving a real-world problem using programming.